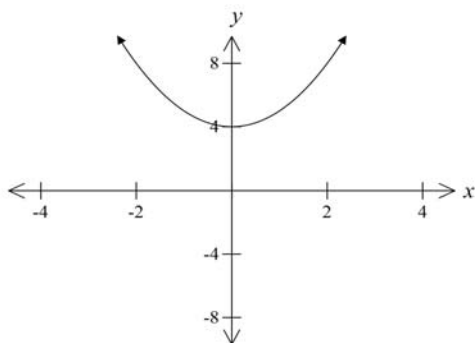


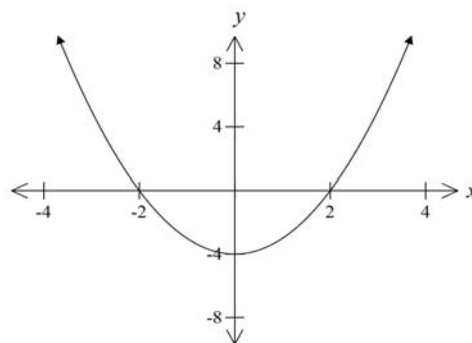
Year 12 Mathematics Ext. 2
Trial Examination 2012
Section 1 Objective Response Questions

- 1 Given that $f(x) = 4 - x^2$, which one of the following graphs best fits the graph of $y = |f(x)|$?

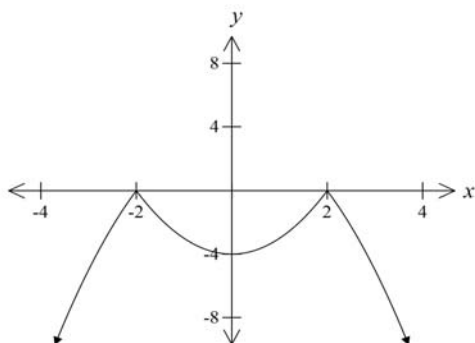
(A)



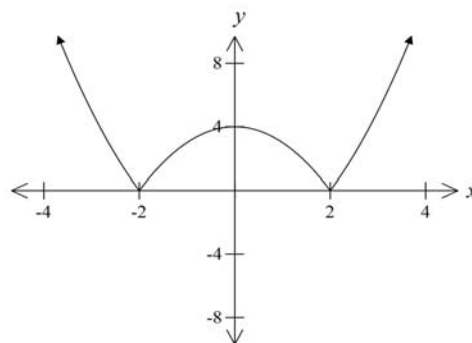
(B)



(C)



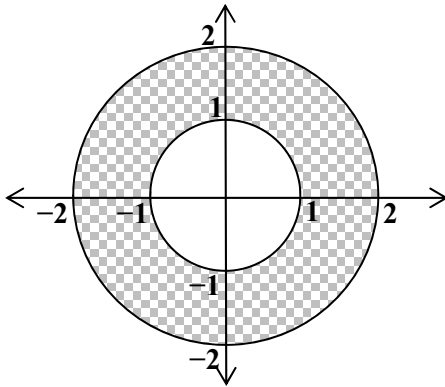
(D)



- 2 It is given that $3 + i$ is a root of $P(z) = z^3 + az^2 + bz + 10$ where a and b are real numbers. Which expression factorises $P(z)$ over the real numbers?

- (A) $(z - 1)(z^2 + 6z - 10)$
 (B) $(z - 1)(z^2 - 6z - 10)$
 (C) $(z + 1)(z^2 + 6z + 10)$
 (D) $(z + 1)(z^2 - 6z + 10)$

3 Consider the Argand diagram below.



Which inequality best defines the shaded area?

- (A) $0 \leq |z| \leq 2$
- (B) $1 \leq |z| \leq 2$
- (C) $0 \leq |z-1| \leq 2$
- (D) $1 \leq |z-1| \leq 2$

4 The points $P(a \cos \theta, b \sin \theta)$ and $Q(a \cos \phi, b \sin \phi)$ lie on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the chord PQ subtends a right angle at $(0,0)$. Which of the following is the correct equation?

- (A) $\tan \theta \tan \phi = -\frac{b^2}{a^2}$
- (B) $\tan \theta \tan \phi = -\frac{a^2}{b^2}$
- (C) $\tan \theta \tan \phi = \frac{b^2}{a^2}$
- (D) $\tan \theta \tan \phi = \frac{a^2}{b^2}$

5 Consider the hyperbola with the equation $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

What are the coordinates of the foci of this hyperbola?

- | | |
|------------------|------------------|
| (A) $(\pm 4, 0)$ | (B) $(0, \pm 4)$ |
| (C) $(0, \pm 5)$ | (D) $(\pm 5, 0)$ |

6 Which one of the following is an expression for $\int \frac{1}{\sqrt{x^2 - 6x + 10}} dx$?

(A) $\ln(x - 3 - \sqrt{x^2 - 6x + 10}) + c$

(B) $\ln(x + 3 - \sqrt{x^2 - 6x + 10}) + c$

(C) $\ln(x - 3 + \sqrt{x^2 - 6x + 10}) + c$

(D) $\ln(x + 3 + \sqrt{x^2 - 6x + 10}) + c$

7 The area bounded by $y = x^3$, the line $y = 8$, and the y axis, is rotated about the y axis to form a solid. The volume of this solid is:

(A) $\frac{2\pi}{5}$ cubic units

(B) $\frac{3\pi}{5}$ cubic units

(C) $\frac{93\pi}{5}$ cubic units

(D) $\frac{96\pi}{5}$ cubic units

8 A particle of mass m falls from rest under gravity and the resistance to its motion is mkv^2 , where v is its speed and k is a positive constant. Which of the following is the correct expression for square of the velocity where x is the distance of the particle fallen from rest?

(A) $v^2 = \frac{g}{k}(1 - e^{-2kx})$

(B) $v^2 = \frac{g}{k}(1 + e^{-2kx})$

(C) $v^2 = \frac{g}{k}(1 - e^{2kx})$

(D) $v^2 = \frac{g}{k}(1 + e^{2kx})$

9 Let α , β and γ be the roots of the equation $x^3 + 3x^2 + 4 = 0$. Which one of the following polynomial equations have roots α^2 , β^2 and γ^2 ?

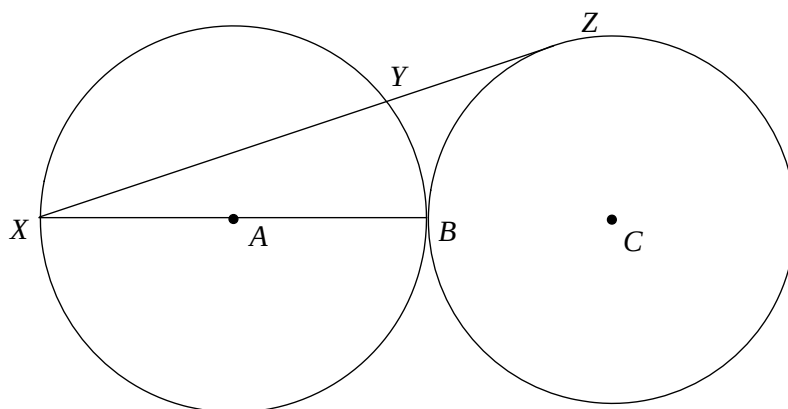
(A) $x^3 - 9x^2 - 24x - 4 = 0$

(B) $x^3 - 3x^2 - 12x - 4 = 0$

(C) $x^3 - 9x^2 - 24x - 16 = 0$

(D) $x^3 - 3x^2 - 12x - 16 = 0$

10



The diagram above shows two circles of equal radii with centres A and C respectively. The two circles touch externally at B and the line XB is a diameter. The line XZ is the tangent to the circle centre C , at Z , cutting the circle, centre A , in Y . Which is the correct expression that relates the length of XZ to the length of XY ?

(A) $3XZ = 4XY$

(B) $XZ = 2XY$

(C) $2XZ = 3XY$

(D) $2XZ = 5XY$

SECTION II Extended Response Questions

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question

Question 1 (15 marks)**Marks**

a) Find $\int \frac{x}{\sqrt{2-x^2}} dx$ using the substitution $x = \sqrt{2} \sin \theta$. **3**

b) i) Find the real numbers a and b such that

$$\frac{1}{x(2x+1)} = \frac{a}{x} + \frac{b}{2x+1}.$$

1

ii) Hence, evaluate $\int_{\frac{1}{2}}^1 \frac{dx}{x(2x+1)}$. **2**

c) Use integration by parts to show that $\int_0^1 \tan^{-1} x \, dx = \frac{\pi}{4} - \frac{1}{2} \ln 2$. **3**

d) Given that $I_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$, where $n \geq 0$ is an integer.

i) Prove that $I_n = \frac{(n-1)}{n} I_{n-2}$, for $n \geq 2$. **4**

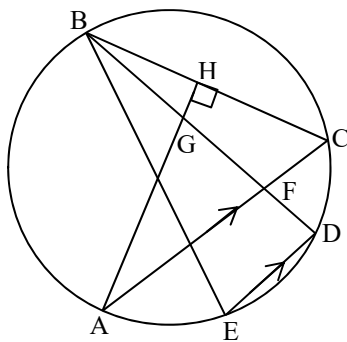
ii) Hence, evaluate $\int_0^{\frac{\pi}{2}} \cos^6 x \, dx$. **2**

Question 2 Begin in a new booklet (15 marks)**Marks**

- a) The complex number v has modulus 1 and argument $\frac{\pi}{6}$, and the complex number w has modulus 2 and argument $\frac{-2\pi}{3}$.
- i) Express wv and iv in the modulus-argument form where each argument is between $-\pi$ and π . **2**
- ii) Show that v is a solution of the equation $Z^4 = iZ$. Hence, or otherwise, state the other two non-zero roots of this equation. You may leave your answers in modulus-argument form. **3**
- iii) Mark, on an Argand diagram, the points P , Q , R and S representing v , w , wv and iv respectively. **1**
- iv) Hence, or otherwise, show that PS is parallel to RQ . **2**
- v) Hence, or otherwise, find a real number u such that $iv - v = u(w - wv)$. **2**
- b) Let $f(x) = \frac{11-x}{x^2-x-2}$,
- i) Draw a one-third page sketch of the graph of $y = f(x)$ showing clearly all asymptotes. (Do not calculate co-ordinates of any turning points) **3**
- ii) Hence or otherwise, draw a one- third page sketch of the graph of $y = \frac{1}{f(x)}$ showing all asymptotes. (Do not calculate the co-ordinates of any turning points) **2**

Question 3 Begin in a new booklet (15 marks)**Marks**

a)



**Diagram
not to
scale**

The diagram shows the points A, B, C, D and E on a circle, such that BE is a diameter and AC is parallel to ED . Also, AH is perpendicular to BC , and BD intersects AH and AC at G and F respectively.

i) Copy the diagram into your answer booklet.

ii) Prove angle BFC is 90° .

2

iii) Prove $CFGH$ is a cyclic quadrilateral.

2

iv) Hence, or otherwise, show that $AB \times BG = BE \times BH$.

3

b) The hyperbola, H , has the equation $\frac{x^2}{25} - \frac{y^2}{9} = 1$.

i) Find the eccentricity of H .

1

ii) Find the co-ordinates of the foci of H .

1

iii) Draw a neat, one-third page sketch of H .

2

iv) The line $x = 6$ cuts H at A and B . Find the co-ordinates of A and B , if A is in the first quadrant.

2

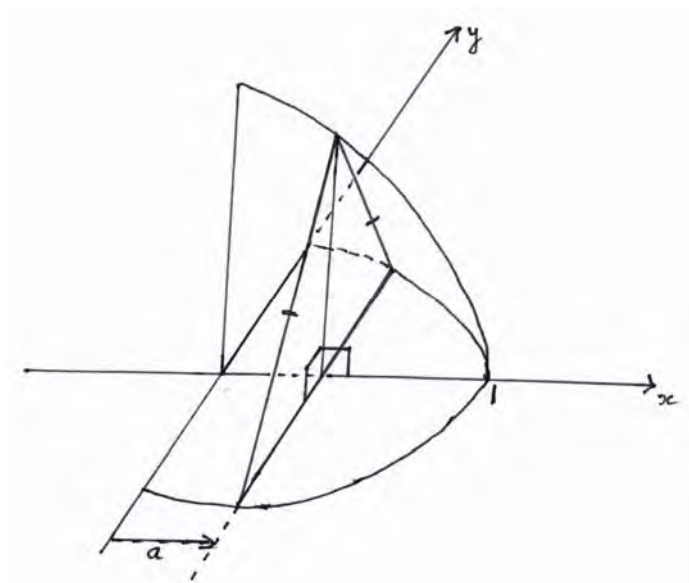
v) Derive the equation of the tangent to H at A .

2

Question 4 Begin in a new booklet (15 marks)**Marks**

- a) If p, q and r are the roots of the equation $x^3 + 4x^2 - 3x + 1 = 0$, find the polynomial equation whose roots are $\frac{1}{p}, \frac{1}{q}$ and $\frac{1}{r}$. **2**
- b) i) Let k be a zero of the polynomial $F(x)$ and also of its derivative $F'(x)$.
Prove that k is a zero of $F(x)$ of multiplicity at least 2. **3**
- ii) Show that $y = 1$ is a root of multiplicity at least 2, of the equation $y^{2t} - ty^{t+1} = 1 - ty^{t-1}$, where $t \geq 2$ is a positive integer. **2**
- c) The polynomial $P(x)$ gives remainders 1 and -2 when divided by $2x - 1$ and $x - 2$ respectively. What is the remainder when $P(x)$ is divided by $2x^2 - 5x + 2$? **3**

d)



**Diagram
not to
scale**

The base of a solid is formed by the area bounded by $x^2 + y^2 = 1$ for $0 \leq x \leq 1$ as shown in the diagram above. Vertical cross sections of the solid taken parallel to the y -axis are in the shape of isosceles triangles with the two equal sides being of length three-quarters the length of the third side which is in the base of the solid.

- i) Show that the area of the triangular cross-section at $x = a$, is $\frac{\sqrt{5}}{2}(1 - a^2)$. **3**
- ii) Hence or otherwise find the volume of the solid. **2**

Question 5 Begin in a new booklet (15 marks)**Marks**

- a) Prove by mathematical induction that, for integers
- $n \geq 2$
- ,

$$\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{3^2}\right)\left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n} . \quad \mathbf{4}$$

- b) i) If
- $\theta = \tan^{-1} A + \tan^{-1} B$
- , show that
- $\tan \theta = \frac{A+B}{1-AB}$
- .
- 1**

- ii) Hence solve the equation
- $\tan^{-1} 3x + \tan^{-1} 2x = \frac{\pi}{4}$
- .
- 3**

- c) i) Sketch the graph of the function
- $y = \sin^{-1}\left(\frac{x}{2}\right)$
- .
- 1**

- ii) Show that the equation of the tangent,
- l
- , to the curve
- $y = \sin^{-1}\left(\frac{x}{2}\right)$
- at the point where
- $x = \sqrt{3}$
- is
- $y = x + \frac{\pi}{3} - \sqrt{3}$
- .
- 2**

- iii) The region, where
- $x \geq 0$
- , bounded by
- $y = \sin^{-1}\left(\frac{x}{2}\right)$
- , the
- y
- axis and the line
- l
- is rotated about the
- y
- axis to form a solid of volume
- V
- .

$$1) \text{ Show } V = \pi\sqrt{3} - 4\pi \int_0^{\frac{\pi}{3}} \sin^2 y \, dy . \quad \mathbf{2}$$

- 2) Hence, or otherwise, find
- V
- .
- 2**

Question 6 Begin in a new booklet (15 marks)**Marks**

- a) A body is projected vertically upwards, under gravity, from the ground in a medium that produces a resistance force per unit mass of kv^2 , where v is the velocity and k is a positive constant.
The acceleration due to gravity is g .

- i) If the initial velocity of the body is v_0 , prove that the maximum height, H , of the body above the ground is given by

$$H = \frac{1}{2k} \log_e \left(1 + \frac{kv_0^2}{g} \right). \quad 4$$

- ii) In a second projection vertically upwards of the body, it is noticed that the maximum height reached is $2H$. Show that the initial velocity was $(e^{2kH} + 1)^{\frac{1}{2}} v_0$. 3

- b) A body is moving in a horizontal straight line. At time t seconds, its displacement is x metres from a fixed point O on the line, and its acceleration is $\frac{-1}{10} \sqrt{v}(1 + \sqrt{v})$ where $v \geq 0$ is its velocity.

The body is initially at O with velocity $V > 0$.

- i) Show that $t = 20 \log_e \left(\frac{1 + \sqrt{V}}{1 + \sqrt{v}} \right)$. 3

- ii) Hence, or otherwise, prove that the body comes to rest. 1

- iii) Find the distance travelled before the body comes to rest. 4

END OF ASSESSMENT

Solution to Year 12 Ext 2 Trial Examination**SECTION 1 MCQ**

1. D
2. D
3. B
4. B
5. D
6. C
7. D
8. A
9. C
10. C

SECTION 11 EXTENDED RESPONSES**Question 1**

$$a) \int \frac{x}{\sqrt{2-x^2}} dx \quad x = \sqrt{2} \sin \theta$$

$$\therefore dx = \sqrt{2} \cos \theta$$

$$= \int \frac{\sqrt{2} \sin \theta \cdot \sqrt{2} \cos \theta}{\sqrt{2 - (\sqrt{2} \sin \theta)^2}} d\theta$$

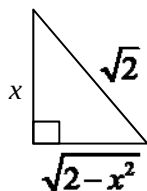
$$= \int \frac{\sqrt{2} \sin \theta \cos \theta}{\cos \theta} d\theta$$

$$= \int \sqrt{2} \sin \theta d\theta$$

$$= -\sqrt{2} \cos \theta + c$$

$$= -\sqrt{2} \cdot \frac{\sqrt{2-x^2}}{\sqrt{2}} + c$$

$$= -\sqrt{2-x^2} + c$$



$$c) \int_0^1 \tan^{-1} x dx$$

$$= \int_0^1 \frac{d(x)}{dx} \cdot \tan^{-1} x dx$$

$$= \left[x \tan^{-1} x \right]_0^1 - \int_0^1 x \cdot \frac{1}{1+x^2} dx$$

$$= \left(\frac{\pi}{4} - 0 \right) - \frac{1}{2} \int_0^1 \frac{2x}{1+x^2} dx$$

$$= \frac{\pi}{4} - \frac{1}{2} \left[\ln(1+x^2) \right]_0^1$$

$$= \frac{\pi}{4} - \frac{1}{2} [\ln 2 - \ln 1]$$

$$= \frac{\pi}{4} - \frac{1}{2} \ln 2$$

Question 1 conti

$$1 \text{ bii) } \frac{1}{x(2x+1)} = \frac{a}{x} + \frac{b}{2x+1}$$

$$a(2x+1) + bx \equiv 1$$

$$\text{Put } x = 0 \Rightarrow a = 1$$

$$\text{Put } x = \frac{-1}{2} \Rightarrow b = -2$$

$$\therefore \frac{1}{x(2x+1)} = \frac{1}{x} - \frac{2}{2x+1}$$

$$\text{bii) } \int_{\frac{1}{2}}^1 \frac{dx}{x(2x+1)}$$

$$= \int_{\frac{1}{2}}^1 \frac{dx}{x} - \int_{\frac{1}{2}}^1 \frac{2dx}{(2x+1)}$$

$$= \left[\ln|x| - \ln|2x+1| \right]_{\frac{1}{2}}^1$$

$$= \left[\ln \left| \frac{x}{2x+1} \right| \right]_{\frac{1}{2}}^1$$

$$= \ln \frac{1}{3} - \ln \left| \frac{1}{2} \right|$$

$$= \ln \frac{4}{3}$$

Question 1 continued

$$di) I_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$$

$$= \int_0^{\frac{\pi}{2}} \cos x \cos^{n-1} x \, dx$$

$$= \left[\sin x \cos^{n-1} x \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} (n-1) \cos^{n-2} x \sin x \cdot \sin x \, dx$$

$$= (0-0) + (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-2} x \sin^2 x \, dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-2} x \, dx - (n-1) \int_0^{\frac{\pi}{2}} \cos^n x \, dx$$

$$I_n = (n-1) I_{n-2} - (n-1) I_n$$

$$I_n + (n-1) I_n = (n-1) I_{n-2}$$

$$n I_n = (n-1) I_{n-2}$$

$$\therefore I_n = \frac{(n-1)}{n} I_{n-2}$$

$$dii) I_6 = \int_0^{\frac{\pi}{2}} \cos^6 x \, dx$$

$$= \frac{5}{6} I_4$$

$$= \frac{5}{6} \cdot \frac{3}{4} \cdot \frac{1}{2} I_2$$

$$= \frac{15}{48} \int_0^{\frac{\pi}{2}} dx$$

$$= \frac{15}{48} \left[x^0 \right]_0^{\frac{\pi}{2}}$$

$$= \frac{15\pi}{96} = \frac{5\pi}{32}$$

Question 2

$$2ai) \quad wv = 2 \left[\cos\left(-\frac{\pi}{2}\right) + i \sin\left(-\frac{\pi}{2}\right) \right]$$

$$iv = \cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right)$$

$$a ii) \quad v = \cos\frac{\pi}{6} + i \sin\frac{\pi}{6} = z$$

$$z^4 = \cos\frac{\pi}{6} \times 4 + i \sin\frac{\pi}{6} \times 4$$

$$= \cos\left(\frac{2\pi}{3}\right) + i \sin\left(\frac{2\pi}{3}\right) = iv$$

$$= iz$$

$\therefore v$ is a solution to $z^4 = iz$

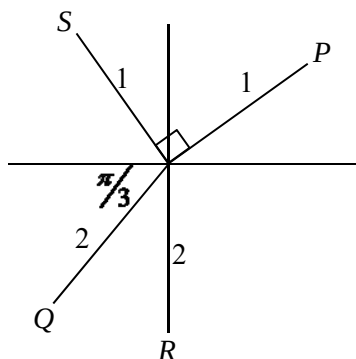
if $z \neq 0$, then $z^3 = i$

non zero roots are $\cos\frac{\pi}{6} + i \sin\frac{\pi}{6} = \frac{1}{2}(\sqrt{3} + i)$

$$\cos\frac{5\pi}{6} + i \sin\frac{5\pi}{6} = \frac{1}{2}(-\sqrt{3} + i) = v \text{cis}\left(\frac{2\pi}{3}\right)$$

and $\cos\left(-\frac{\pi}{2}\right) + i \sin\left(-\frac{\pi}{2}\right) = -i = \frac{wv}{2}$

a iii)



Question 2 conti

aiv) Method 1

$$\text{In } \triangle OQR \quad x = \frac{1}{2}(180^\circ - 30^\circ) = 75^\circ$$

$$\text{In } \triangle OSP \quad x = [180^\circ - 45^\circ - (90^\circ - 30^\circ)] = 75^\circ$$

$$\therefore \overline{PS} \parallel \overline{RQ}$$

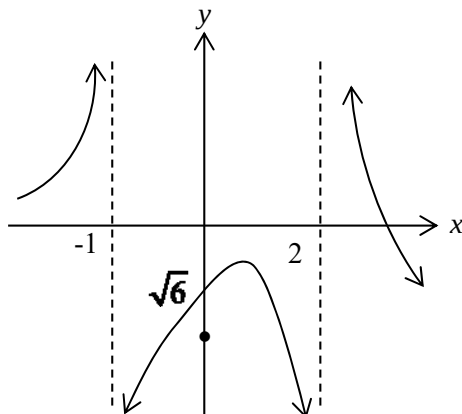
Method 2

Show $\overline{PS} = \lambda \overline{RQ}$ where $\lambda \in \text{reals}$

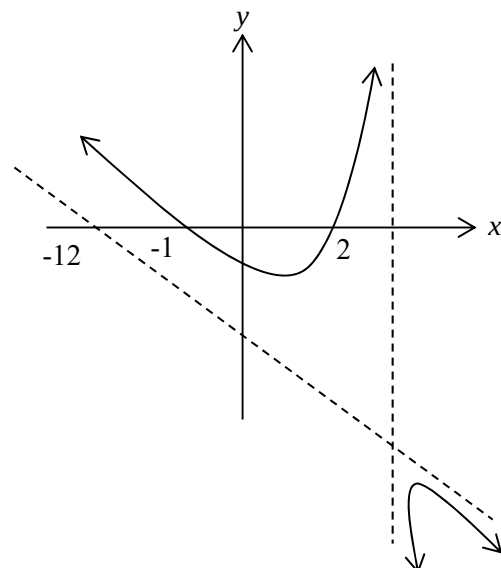
$$\begin{aligned} \overline{RQ} &= w - wv \\ &= 2\text{cis}\left(-\frac{2\pi}{3}\right) - (2i) \\ &= \frac{2}{2}[-1 - i\sqrt{3}] + 2i \\ &= -1 + i[2 - \sqrt{3}] \end{aligned}$$

$$\begin{aligned} \overline{PS} &= iv - v \\ &= \text{cis}\left(\frac{2\pi}{3}\right) - \text{cis}\left(\frac{\pi}{6}\right) \\ &= \frac{1}{2}[-1 + i\sqrt{3}] - \frac{1}{2}[\sqrt{3} + i] \\ &= 2 \frac{(\sqrt{3} + 1)}{2} \times \overline{RQ} \quad \text{by inspection note } (\sqrt{3} + 1) \times (2 - \sqrt{3}) = \sqrt{3} - 1 \\ &\Rightarrow \overline{PS} \parallel \overline{RQ} \end{aligned}$$

bi)



bii)



Question 3

3a) $\angle EDB = 90^\circ$ (angle in a semi circle)
 $\angle BFC = 90^\circ$ (alt angles, $AC \parallel ED$, suppl angles)

iii) $\angle GHC + \angle GFC = 90 + 90 = 180^\circ$
 $\therefore CFGH$ is a cyclic quad (opps angles are suppl)

iv) In $\triangle BGH$, $\triangle BEA$

$\angle BGH = \angle BAE = 90$ (diameter subtends right angle at cfce)

$\angle BCA = \angle BEA$ (angles subtended by a common chord BA)

but $\angle BCA = \angle BGH$ (exterior angle of cyclic quad equal to int opp anlgcs)

$\therefore \angle BGH = \angle BEA$

$\therefore \triangle BGH \equiv \triangle BEA$ (equi-angular)

$\therefore \frac{BG}{BE} = \frac{BH}{AB}$ (corresp sides of similar Δ)

$\therefore AB \cdot BG = BE \cdot BH$

3bi) $a = 5, b = 3$

$$b^2 = a^2(e^2 - 1)$$

$$9 = 25(e^2 - 1)$$

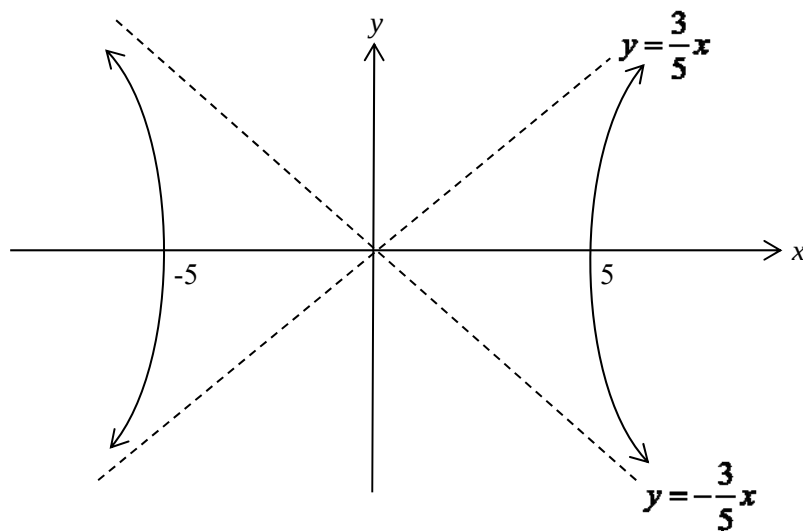
$$e^2 = \frac{34}{25}$$

$$e = \frac{\sqrt{34}}{5}$$

ii) Foci $S(ae, 0) = (\sqrt{34}, 0)$

$$S'(-ae, 0) = (-\sqrt{34}, 0)$$

iii)



Question 3 conti

$$3biv) \frac{36}{25} - \frac{y^2}{9} = 1$$

$$y^2 = \frac{9 \times 11}{25}$$

$$y = \frac{\pm 3\sqrt{11}}{5}$$

$$A\left(6, \frac{3\sqrt{11}}{5}\right) \quad B\left(-6, \frac{-3\sqrt{11}}{5}\right)$$

$$3bv) \frac{x^2}{25} - \frac{y^2}{9} = 1$$

$$\frac{2x}{25} - \frac{2y}{9} \cdot \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{2x}{25} \cdot \frac{9}{2y}$$

$$\therefore \frac{dy}{dx} = \frac{9x}{25y}$$

$$\text{at } \left(6, \frac{99}{5}\right), \frac{dy}{dx} = \frac{54}{25 \cdot \sqrt{99}}$$

$$\therefore \text{equation is } y - \frac{3\sqrt{11}}{5} = \frac{18}{5\sqrt{11}}(x - 6)$$

$$y = \frac{18}{5\sqrt{11}}x - 6 + \frac{3\sqrt{11}}{5}$$

Question 4

4a) $x^3 + 4x^2 - 3x + 1 = 0$ has no root $x = 0$.

$$\text{let } y = \frac{1}{x} \text{ and } x = \frac{1}{y}$$

the equation below has roots

$$\frac{1}{p}, \frac{1}{q}, \frac{1}{r}$$

$$\frac{1}{y^3} + \frac{4}{y^2} - \frac{3}{y} + 1 = 0$$

$$y^3 - 3y^2 + 4y + 1 = 0$$

bii) k is zero of $F(x)$

$$\Rightarrow F(x) = (x-4)q(x)$$

$$F'(x) = (x-k)q'(x) + q(x)$$

k is zero of $F'(x)$

$$F'(k) = 0$$

$$= (k-k)q'(k) + q(k) = 0$$

$$\Rightarrow q(k) = 0$$

$$\Rightarrow q(k) = (x-4)r(x)$$

$$\text{hence } F(x) = (x-4)^2 r(x)$$

$\Rightarrow k$ is at least a root of multiplicity 2

bii) $y^{2t} - ty^{t+1} = 1 - ty^{t-1}$

$$t = 1, 2, 3, \dots \Rightarrow y^{2t} - ty^{t+1} - 1 + ty^{t-1} = 0$$

$$\text{let } P(y) = \text{LHS of above} \therefore P(1) = 1 - t \times 1 + t \times 1 - 1 = 0$$

$$\therefore y = 1 \text{ is a root of } P(y) = 0$$

$$P'(y) = 2ty^{2t-1} - t(t+1)y^t + t(t-1)y^{t-2}$$

$$\text{for } t = 2, 3, 4, \dots \Rightarrow P'(1) = 2t - t^2 - t + t^2 - t = 0$$

$\therefore$ need to consider the case where $t = 1$ to complete the proof

$$P(y) = y^2 - y^2 + 1 \times y^0 - 1$$

$$\text{So } t = 1 \Rightarrow P(y) = y^2 - y^2 + 1 - 1 \equiv 0 \quad [\text{omit this case } t \geq 2]$$

Question 4 conti

$$4c) P(x) = (2x^2 - 5x + 2)Q(x) + R(x)$$

$$\text{Since degree } D(x) > \deg R(x)$$

$$\deg R(x) < 2$$

$$\text{let } R(x) = ax + b$$

$$\therefore P(x) = (2x - 1)(x - 2)Q(x) + ax + b$$

$$P\left(\frac{1}{2}\right) = \frac{a}{2} + b = 1$$

$$P(2) = 2a + b = -2$$

$$-3b = -6$$

$$b = 2, a = -2$$

$$\therefore R(x) = -2x + 2$$

$$4d) \text{ when } x = a, y = 2\sqrt{1-a^2}$$

$$\therefore \text{length of base} = 2\sqrt{1-a^2}$$

$$h^2 = \left(\frac{3}{2}\sqrt{1-a^2}\right)^2 - \left(\sqrt{1-a^2}\right)^2$$

$$= \frac{a}{4}(1-a^2) - (1-a^2)$$

$$= \frac{5}{4}(1-a^2)$$

$$h = \frac{\sqrt{5(1-a^2)}}{2}, h > 0$$

Question 5**5 (a)** Prove true for $n = 2$

$$\begin{aligned}
 1. \quad LHS &= 1 - \frac{1}{2} \times 2 & RHS &= \frac{2+1}{2 \times 2} \\
 &= \frac{3}{4} & &= \frac{3}{4} \\
 \therefore \text{True form } &= 2
 \end{aligned}$$

2. Assume true form $n = k$

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{k^2}\right) = \left(\frac{k+1}{2k}\right)$$

Step 3. Prove true for $n = k+1$

$$\left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{k^2}\right) \left(1 - \frac{1}{(k+1)^2}\right) = \frac{k+1+1}{2(k+1)}$$

$$LHS \frac{k+1}{2k} \left(1 - \frac{1}{(k+1)^2}\right) \text{ from } \textcircled{2}$$

$$= \frac{k+1}{2k}, \frac{(k+1)^2 - 1}{(k+1)^2}$$

$$= \frac{k+1}{2k}, \frac{k^2 + 2k + 1 - 1}{(k+1)^2}$$

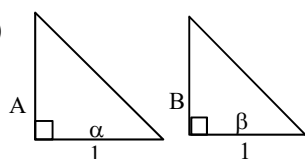
$$= \frac{(k+1)}{2k}, \frac{k(k+2)}{k+1^2}$$

$$= \frac{k+1+1}{2(k+1)}$$

$$= RHS$$

$$\therefore \text{true for } n = k+1$$

$$\therefore \text{true by MI}$$

5 b(i)

$$\text{Let } \tan^{-1} A = \alpha$$

$$\tan^{-1} B = \beta$$

$$\theta = \tan^{-1} \beta + \tan^{-1} A$$

$$= \alpha + \beta$$

$$\tan \theta = \tan(\alpha + \beta)$$

$$= \frac{\tan \alpha \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\therefore \tan \theta = \frac{A+B}{1-AB}$$

Question 5 conti**(ii) from (i)**

$$\tan \frac{\pi}{4} = \frac{3x+2x}{1-3x_1 2x} \Rightarrow 1 = \frac{5x}{1-6x^2}$$

$$\therefore 6x^2 + 5x - 1 = 0$$

$$(6x-1)(x+1) = 0$$

$$x = \frac{1}{6}, -1$$

$$x = \frac{1}{6}$$

$$\text{Test: } \tan \frac{\pi}{4} = \frac{3 \times \frac{1}{6} + 2 \times \frac{1}{6}}{1 - 3 \times \frac{1}{6} \times 2 \times \frac{1}{6}} \\ = 1 \quad \therefore \text{True}$$

Test $x = -1$

$$\tan \frac{\pi}{4} = \frac{3 \times -1 + 2 \times -1}{1 - 3 \times -1 \times 2 \times -1}$$

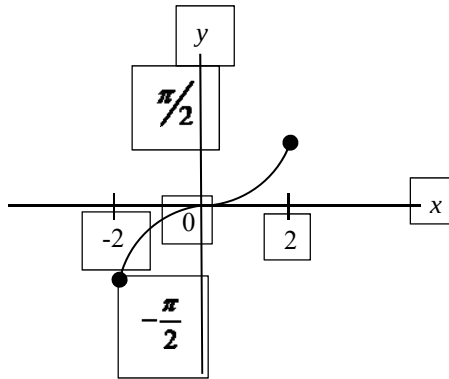
$$= -1 \quad \therefore \text{not true}$$

 $\therefore$ Solution

$$x = \frac{1}{6}$$

Question5 cont

c(i) $y = \sin^{-1} \frac{x}{2}$



(ii) $y' = \frac{1}{2} \times \frac{1}{\sqrt{\frac{4-x^2}{4}}}$

When $x = \sqrt{3}$

$y' = 1$

$x = \sqrt{3}, y = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3}$

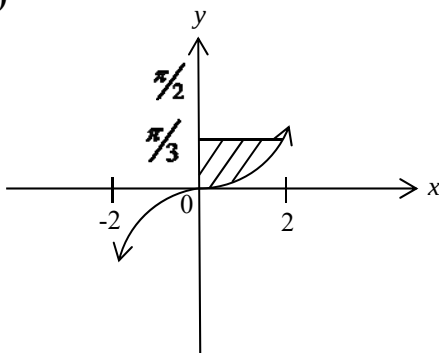
Equation tan:

$y - \frac{\pi}{3} = 1(x - \sqrt{3})$

$y = x - \sqrt{3} + \frac{\pi}{3}$

$\therefore y \text{ int } \left(\frac{\pi}{3}, -\sqrt{3}\right)$

(iii)



$$\Delta V = \left(\frac{-\pi}{3} + \sqrt{3} \right) - \pi x^2 \Delta y \quad \sin y = \frac{x}{2} \therefore x = 2 \sin y$$

$$\Delta V = \left(\left(\frac{-\pi}{3} + \sqrt{3} \right) - \pi (2 \sin y)^2 \right) \Delta y$$

$$V = \lim_{\Delta V \rightarrow 0} \sum_{y=0}^{\frac{\pi}{3}} \left(\left(\frac{-\pi}{3} + \sqrt{3} \right) - \pi (2 \sin y)^2 \right) \Delta y$$

$$= \int_0^{\frac{\pi}{3}} \left(\frac{-\pi}{3} + \sqrt{3} \right) - \pi (2 \sin y)^2 dy$$

=

5c (iii) 1)

$$y = \sin^{-1} \left(\frac{x}{2} \right)$$

$$\frac{x}{2} = \sin y$$

$$x^2 = 4 \sin^2 y$$

$$V = \int_0^{\frac{\pi}{3}} 4 \sin^2 y dy$$

So Volume = Vol of cone - vol of shaded area about y- axis

$$\therefore V = \frac{1}{3} \pi (\sqrt{3})^2 \left[\frac{\pi}{3} - \left(\frac{\pi}{3} - \sqrt{3} \right) \right] - \int_0^{\frac{\pi}{3}} 4 \sin^2 y dy$$

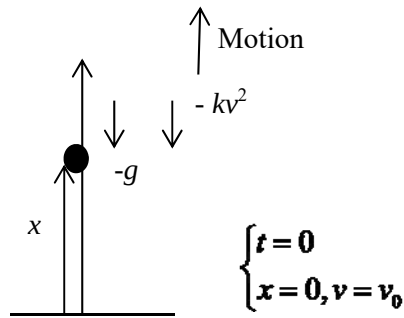
$$V = \pi \sqrt{3} - 4\pi \int_0^{\frac{\pi}{3}} \sin^2 y dy$$

5 c(iii) 2)

$$\begin{aligned} V &= \pi\sqrt{3} - 4\pi \int_0^{\frac{\pi}{3}} (1 - \cos 2y) dy \\ &= \pi\sqrt{3} - 2\pi \left[y - \frac{1}{2} \sin 2y \right]_0^{\frac{\pi}{3}} \\ &= \pi\sqrt{3} - 2\pi \left[\frac{\pi}{3} - \frac{1}{2} \sin \frac{2\pi}{3} \right] \\ &= \pi\sqrt{3} - \frac{2\pi^2}{3} + \frac{\pi\sqrt{3}}{2} \\ &= \frac{\pi 3\sqrt{3}}{2} - \frac{2\pi^2}{3} u^3 \end{aligned}$$

Question 6

(a)



$$(i) \quad \sum F = -(kv^2 + g)$$

$$\frac{dv}{dt} = -(kv^2 + g)$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -(kv^2 + g)$$

$$v \frac{dv}{dx} = -(kv^2 + g)$$

$$\int \frac{v dv}{kv^2 + g} = -\int dx$$

$$\frac{1}{2k} \ln(kv^2 + g) = -x + c$$

$$\ln(kv^2 + g) = -2kx + c$$

$$x=0 \quad v=v_0$$

$$\text{So } c = \ln[kv_0^2 + g]$$

$$\text{At } x=H, v=0$$

$$\ln(g) = -2kH + \ln(kv_0^2 + g)$$

$$= \ln \left(1 + \frac{kv_0^2}{g} \right)$$

$$\therefore H = \frac{1}{2k} \log_e \left(1 + \frac{kv_0^2}{g} \right)$$

Alternate solution to **6ai)**

$$F = -mg - mkv^2$$

$$ma = -mg - mkv^2$$

$$a = -g - kv^2$$

$$\frac{v dv}{dx} = -g - kv^2$$

$$\int_{V_0}^0 \frac{v dv}{g + kv^2} = \int_0^H dx$$

$$\frac{1}{2k} \left[\ln |g + kv^2| \right]_{V_0}^0 = [x]_0^H$$

$$\frac{1}{2k} \left[\ln |g| - \ln |g + kv^2| \right] = -H$$

$$\therefore H = \frac{1}{2k} \left[\ln \frac{g + kv^2}{g} \right], \therefore H = \frac{1}{2k} \ln \left| 1 + \frac{kv_0^2}{g} \right|$$

(6aii)

$$H = \frac{1}{2k} \ln \left[1 + \frac{kv_0^2}{g} \right]$$

$$\Rightarrow \frac{kv_0^2}{g} = e^{2kH} - 1$$

Let V_0 be the initial velocity in the second projection.

$$\therefore \frac{kv_0^2}{g} = e^{4kh} - 1$$

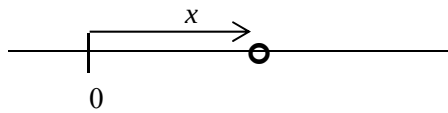
$$\begin{aligned}\frac{V_0^2}{v_0^2} &= \frac{e^{kh} - 1}{e^{2kh} - 1} \\ &= \frac{(e^{2kh} + 1)(e^{2kh} - 1)}{(e^{2kh} - 1)} \\ &= e^{2kh} + 1\end{aligned}$$

$$\therefore \left(\frac{V_0}{v_0}\right) = (e^{2kh} + 1)$$

by $V_0 > 0, v_0 > 0$

$$V_0 = (e^{2kh} + 1)^{\frac{1}{2}} v_0$$

Question 6 (b)



$$t = 0 \quad x = 0, v = V > 0$$

$$(i) \quad \frac{dv}{dt} = -\frac{\sqrt{v}}{10}(1 + \sqrt{v})$$

$$\int \frac{2dv}{2\sqrt{v}(1 + \sqrt{v})} = -\int \frac{dt}{10}$$

$$2\ln(1 + \sqrt{v}) = -\frac{t}{10} + c$$

$$20\ln(1 + \sqrt{v}) = -t + c \quad \text{at } t = 0, v = V > 0$$

$$c = 20\ln(1 + \sqrt{v})$$

$$\therefore t = 20 \geq \left[\ln(1 + \sqrt{v}) - \ln(1 + \sqrt{V}) \right]$$

$$= 20\ln\left(\frac{1 + \sqrt{V}}{1 + \sqrt{v}}\right)$$

$$6 \text{ (bii)} \quad t = 20\ln\left(\frac{1 + \sqrt{V}}{1 + \sqrt{v}}\right)$$

When $v = 0$ $t = 20\ln(1 + \sqrt{V})$ so particle comes to rest (as $t > 0$)

$$6 \text{ (biii)} \quad \frac{dv}{dt} = v \quad \frac{dv}{dx} = -\frac{\sqrt{v}}{10}(1 + \sqrt{v})$$

$$\int \frac{v \, dv}{\sqrt{v}(1+\sqrt{v})} = -\frac{1}{10} \int dx$$

$$\int \frac{\sqrt{v} \, dv}{1+\sqrt{v}} = -\frac{x}{10} + c$$

$$LHS = \int \frac{\sqrt{v}}{1+\sqrt{v}} dv$$

$$\text{Let } y = \sqrt{v} > 0$$

$$dy = \frac{1}{2\sqrt{v}} dv$$

$$\therefore dv = 2\sqrt{v} \, dy \\ = 2y \, dy$$

$$\begin{aligned} \therefore LHS &= \int \frac{y}{1+y} \times 2y \, dy \\ &= 2 \int \frac{y^2 - 1 + 1}{(y+1)} \, dy \\ &= 2 \int \left(y - 1 + \frac{1}{y+1} \right) dy \\ &= y^2 - 2y + 2 \ln(y+1) \\ &= (Vv)^2 - 2\sqrt{v} + 2 \ln(\sqrt{v} + 1) \\ &= v - 2\sqrt{v} + 2 \ln(\sqrt{v} + 1) \\ &= v - 2\sqrt{v} + 2 \ln(\sqrt{v} + 1) \\ &= -\frac{x}{10} + c \end{aligned}$$

Where $x = 0$, $v = V > 0$

$$\therefore C = V - 2\sqrt{V} + 2 \ln(\sqrt{V} + 1)$$

Hence $v = 0$ when $x = D$

D = distance travelled before body comes to rest.

$$\frac{D}{10} = C$$

$$\therefore D = 10 \left[V - 2\sqrt{V} + 2 \ln(\sqrt{V} + 1) \right]$$

Remark

$$D(0) = 0$$

$$\frac{d}{dv} D(V) = \frac{\sqrt{v}}{1 + \sqrt{v}} > 0 \text{ for } V > 0$$

$\therefore D(V)$ is an increasing function $\Rightarrow$

$$D(V) \geq 0 \text{ for } V \geq 0$$

End of solutions